

Complex Measures

Srinjena Das

August 2023

Contents

0.1	Preface	5
0.2	Notations	6
I	Convolution and smooth functions	7
1	Convolution	9
1.1	Properties of Convolution	9
1.1.1	Young's inequality	10
1.2	How to use convolution to cook up "nice" classes of functions?	11
1.2.1	L^∞ functions	11
1.2.2	Continuous functions	11
1.2.3	C^1 functions	12
1.2.4	Smooth functions	13
2	Smooth Functions	15
2.1	Mollifiers in $\mathbb{R}^n$	15
2.1.1	Mollification of functions	16
2.1.2	Lebesgue points	17
2.2	Convergence of ϵ -mollification of functions	17
2.2.1	Pointwise convergence in L^1_{loc} functions at Lebesgue points	17
2.2.2	Uniform convergence in continuous functions on compact subsets	19
2.2.3	Convergence of L^p_{loc} functions on compact sets	19
2.3	Partition of Unity	23
2.3.1	Partition of Unity I	23
2.3.2	Partition of Unity II	25
2.3.3	Smooth Urysohn's lemma	27
II	Differentiation Theory	29
3	Signed Measures	31
3.1	Hahn decomposition theorem	33
3.2	Jordan decomposition theorem	34
3.3	Total variation measure of a signed measure	34
3.3.1	Important results	34
3.3.2	Absolute continuity	35
3.4	Radon-Nikodyn theorem	35
3.4.1	The Radon-Nikodyn derivative	36

4	Complex measures	37
4.1	Total variation of a complex measure	37
4.2	The space of complex measures	38

0.1 Preface

This is notes for the one semester course on Complex Measures taking place in IISER KOLKATA. There will be three modules that will be covered in this course:

1. Convolution and smooth functions (It provides us with a general recipe for constructing smooth functions).
2. Differentiation theory (It provides us with a generalisation of Fundamental theorem of Calculus).
3. Fourier series (It tells us when can we write a periodic function in terms of sines and cosines).

In the first module we would actually like to understand how given any continuous function we can construct a smooth function that approximates it. This type of approximation is very important in fields such as differential geometry. Some examples could be that when we try to obtain a smooth analogue of a map which is a homotopy equivalence (which is a continuous function) between two manifolds, in general, topological spaces; or a smooth path which approximates an arbitrary path (which is again a continuous function) in a homotopy class of a fundamental group of a topological space. Understanding the construction of such smooth maps would lead us to the main goal of studying this module, that is to prove that *compactly supported smooth maps are dense in L^p spaces*, which is a generalisation of an earlier result we learnt: compactly supported continuous functions are smooth in L^p spaces. The second module is where the heart of this course lies and speaking of the third module, which is an application of all the results that we will learn before it in this course, we would also need some tools from functional analysis to understand it deeply.

In this note, mostly the theorems or lemmas that we will encounter, a sketch of their proof will be provided, rather than a detailed proof. We would be using that time to solve questions as that always helps in better understanding of a topic at hand.

0.2 Notations

If $\mathbb{R}^n$ is our measure space, then,

1. Lebesgue measure is the function $\mathcal{L}^n(-) : \mathcal{P}(\mathbb{R}^n) \rightarrow [0, \infty)$ defined as

$$\mathcal{L}^n(A) := \inf \left\{ \sum_{m \in \mathbb{N}} |R_m| : A \subset \bigcup_{m \in \mathbb{N}} R_m, R_m \in \text{cell}_0(\mathbb{R}^n), \text{ for all } m \in \mathbb{N} \right\},$$

where

$$\text{cell}_0(\mathbb{R}^n) := \{I_1 \times \cdots \times I_n : I_i \text{ is a bounded interval in } \mathbb{R}\}$$

and

$$|I_1 \times \cdots \times I_n| = \prod_{j=1}^n l(I_j).$$

2. $\mathcal{M}(\mathbb{R}^n)$ will denote the set of all measurable subsets of $\mathbb{R}^n$, that is,

$$\mathcal{M}(\mathbb{R}^n) := \{A \subseteq \mathbb{R}^n : A \text{ is measurable}\}$$

3. $\mathcal{M}_f(\mathbb{R}^n)$ will denote the set of all measurable functions on $\mathbb{R}^n$, that is,

$$\mathcal{M}_f(\mathbb{R}^n) := \{f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}} : f \text{ is a measurable function}\}.$$

Part I

Convolution and smooth functions

Chapter 1

Convolution

Convolution can be thought of as a binary operation between two measurable functions of $\mathbb{R}^n$ which yields a function that is well defined a.e. and have properties much finer than the properties of the initial functions that we considered. Before knowing what properties we are talking about, let us give the precise definition of convolution followed by some examples.

Definition 1.0.0.0.1. (*Convolution*). Let $n \in \mathbb{N}$ and $f, g \in \mathcal{M}_f(\mathbb{R}^n)$. The convolution of f and g denoted by $f * g$ is defined as

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x - y)dy$$

for all $x \in \mathbb{R}^n$, whenever the integral is well defined. (By well definedness of the integral we are referring to the finiteness of f^+, f^-, g^+, g^- as they could be infinite even though the functions f and g are measurable.)

Example 1.0.0.0.2. Let us define $f, g : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = \chi_{[-1,1]}(x)$ and $g(x) = \chi_{[0,2]}(x)$ for all $x \in \mathbb{R}$ and we would like to calculate $f * f, f * g, g * g$.

$$(f * f)(x) = \int_{\mathbb{R}} f(y)f(x - y)dy = \int_{\mathbb{R}} \chi_{[-1,1]}(y)\chi_{[-1,1]}(x - y)dy = \mathcal{L}^1([-1, 1] \cap [x - 1, x + 1]).$$

With some calculations, one can conclude

$$(f * f)(x) = \begin{cases} 2 - |x| & \text{if } |x| < 2 \\ 0 & \text{if } |x| \geq 2. \end{cases}$$

$$\text{Now, } (f * g)(x) = \int_{\mathbb{R}} f(y)g(x - y)dy = \int_{\mathbb{R}} \chi_{[-1,1]}(y)\chi_{[0,2]}(x - y)dy = \mathcal{L}^1([-1, 1] \cap [x - 2, x]).$$

Similarly, we can compute for this case too to get

$$(f * g)(x) = \begin{cases} x + 1 & \text{if } |x| < 1 \\ 3 - x & \text{if } 1 \leq x < 3 \\ 0 & \text{if } x \in (-\infty, -1] \cup [3, \infty). \end{cases}$$

$(g * g)(x)$ can be computed in the same way.

1.1 Properties of Convolution

To know the class of functions where the convolution integral is well defined almost everywhere, we consider the following two theorems:

Theorem 1.1.0.0.1. Let $n \in \mathbb{N}$ and $f, g \in L^1(\mathbb{R}^n)$. Then,

1. $(f * g) \in L^1(\mathbb{R}^n)$ satisfying the estimate $\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}$.

2. $(f * g)$ is finite almost everywhere.

Proof. To prove (1), it is enough to expand the definition of $f * g$ and use Fubini-Tonelli and (2) directly follows from (1) and the fact that every L^1 function is finite almost everywhere. $\square$

1.1.1 Young's inequality

Theorem 1.1.1.0.1. (Young's inequality). Let $n \in \mathbb{N}$ and $p, q, r \in [1, \infty]$ that satisfy

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}. \quad (1.1)$$

Then for all $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, we have $f * g \in L^r(\mathbb{R}^n)$ with

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}. \quad (1.2)$$

Proof. The proof of this theorem is an application of Hölder's inequality in three functions and Fubini-Tonelli. To prove this we divide into three cases: (1) $pq < \infty, r < \infty$, (2) $pq < \infty, r = \infty$, (3) $pq = \infty$.

Case 1: $pq < \infty, r < \infty$.

Let p', q', r' be the conjugate exponents of p, q, r respectively. Then from 1.1 it follows that

$$\frac{1}{p'} + \frac{1}{r} = \frac{1}{q} \quad (1.3)$$

$$\frac{1}{q'} + \frac{1}{r} = \frac{1}{p} \quad (1.4)$$

$$\frac{1}{p'} + \frac{1}{q'} + \frac{1}{r} = 1. \quad (1.5)$$

Expanding the definition of convolution of f and g and using the first two of the above equations, we get

$$|(f * g)(x)| = \int_{\mathbb{R}^n} |f(y)|^{p/q'} |f(y)|^{p/r} |g(x-y)|^{q/p'} |g(x-y)|^{q/r} dy$$

Considering the three functions $|f(y)|^{p/q'}$, $|g(x-y)|^{q/p'}$, $|f(y)|^{p/r} |g(x-y)|^{q/r}$ and using Hölder's inequality, we get

$$|(f * g)(x)| \leq \|f\|_{L^p}^{p/q'} \|g\|_{L^q}^{qr/p'} \left(\int_{\mathbb{R}^n} |f(y)|^p |g(x-y)|^q dy \right)^{1/r}.$$

Raising both sides to the power of r , integrating wrt x and using Fubini-Tonelli, we get

$$\begin{aligned} \|f * g\|_{L^r}^r &\leq \|f\|_{L^p}^{pr/q'} \|g\|_{L^q}^{qr/p'} \|g\|_{L^q}^q \|f\|_{L^p}^p \\ &= \|f\|_{L^p}^r \|g\|_{L^q}^r, \end{aligned}$$

and thus (1.2) follows.

Case 2: $pq < \infty, r = \infty$.

Note that for all $x \in \mathbb{R}^n$,

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)dy.$$

(1.2) directly follows after using Hölder's inequality to the above.

Case 3: $pq = \infty$.

WLOG, let us assume that $p = \infty$. Since, $q \geq 1$, (1.1) leaves us with only one choice of r and q , that is, $q = 1$ and $r = \infty$. In other words, we have to show that

$$\|f * g\|_{L^\infty} \leq \|f\|_{L^\infty} \|g\|_{L^1}.$$

But this is obvious just by expanding the definition of $f * g$ and applying the respective norms to get the above inequality. $\square$

Remark 1.1.1.0.2. 1. When we earlier mentioned that convolution of two functions has better properties than the initial functions in consideration, we meant that convolution improves the order of integrability. It follows from 1.1 that $r \geq \max\{p, q\}$.

2. When p and q are conjugate exponents, we have $r = \infty$, that is, $f * g \in L^\infty(\mathbb{R}^n)$.

3. It is just a matter of check that convolution is commutative, associative and distributive.

1.2 How to use convolution to cook up "nice" classes of functions?

Other than improving the order of integrability, there are other benefits that convolution of two specific classes of function provide, and they are highlighted by the following two theorems:

1.2.1 L^∞ functions

Theorem 1.2.1.0.1. *Let $n \in \mathbb{N}$. If $f \in L^1(\mathbb{R}^n)$ and $g \in L^\infty(\mathbb{R}^n)$, then their convolution $(f * g) \in L^\infty(\mathbb{R}^n)$ is uniformly continuous.*

Proof. For all $x, y \in \mathbb{R}^n$, by definition of convolution, we have that

$$(f * g)(x) - (f * g)(y) = \int_{\mathbb{R}^n} [f(x - z) - f(y - z)]g(z)dz.$$

Then,

$$\begin{aligned} |(f * g)(x) - (f * g)(y)| &\leq \|g\|_\infty \int_{\mathbb{R}^n} |f(x - z) - f(y - z)|dz \\ &= \|g\|_\infty \int_{\mathbb{R}^n} |f(x - y + w) - f(w)|dw \\ &= \|g\|_\infty \|T_{x-y}f - f\|_{L^1(\mathbb{R}^n)}, \end{aligned}$$

where T_{x-y} denotes the translation of f by $(x - y)$. Since, translation is a continuous function and $g \in L^\infty$, for given $x, y \in \mathbb{R}^n$, $\exists \delta > 0$ such that $\|x - y\| < \delta$ implies that $|(f * g)(x) - (f * g)(y)| < \epsilon$. $\square$

1.2.2 Continuous functions

Theorem 1.2.2.0.1. *Let $N \in \mathbb{N}$ and $p, q \in (1, \infty)$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, their convolution $f * g \in C(\mathbb{R}^n)$ satisfying*

$$\lim_{\|x\| \rightarrow \infty} f * g = 0.$$

Proof. Since, compactly supported continuous functions are dense in L^p spaces, we can find two sequences $(f_m)_{m \in \mathbb{N}}$ and $(g_m)_{m \in \mathbb{N}}$ in $C_c(\mathbb{R}^n)$ such that $f_m \rightarrow f$ in $L^p(\mathbb{R}^n)$ and $g_m \rightarrow g$ in $L^q(\mathbb{R}^n)$. —(*)

Note that, since continuous functions are L^1 functions and compactly supported functions are L^∞ functions, it follows from the previous theorem that $f_m * g_m \in C(\mathbb{R}^n)$. Also, $\text{supp}(f_m * g_m)$ is compact, since it is a closed set lying in the intersection of the compact sets $\text{supp } f_m$ and $\text{supp } g_m$. Thus, $f_m * g_m \in C_c(\mathbb{R}^n)$.

Then, for all $x \in \mathbb{R}^n$, first using triangle inequality and then Hölder's, we get

$$|(f_m * g_m)(x) - (f * g)(x)| \leq \|f_m\|_{L^p} \|g_m - g\|_{L^q} + \|f_m - f\|_{L^p} \|g\|_{L^q},$$

which implies

$$\|(f_m * g_m)(x) - (f * g)(x)\|_{L^\infty} \leq \|f_m\|_{L^p} \|g_m - g\|_{L^q} + \|f_m - f\|_{L^p} \|g\|_{L^q}.$$

Using (*), it follows that $f_m * g_m \rightarrow f * g$ uniformly on $\mathbb{R}^n$, and hence, $f * g \in C(\mathbb{R}^n)$. Also, since

$$\overline{C_c(\mathbb{R}^n)}^{\|\cdot\|_{L^\infty}} = \{f \in C(\mathbb{R}^n) \mid \lim_{\|x\| \rightarrow \infty} f(x) = 0\},$$

the result follows. $\square$

Remark 1.2.2.0.2. 1. We cannot expect $f * g$, in theorem 1.2.1.0.1, to satisfy the conclusions of theorem 1.2.2.0.1. This can be easily seen by taking the function g to be identically 1.

2. In theorem 1.2.2.0.1, the limit of $f * g$ vanishing at infinity also implies that $f * g$ is uniformly continuous.

3. The above two theorems give us a way to come up with functions which are uniformly continuous.

1.2.3 C^1 functions

It is obvious to ask, if we can cook up functions which are uniformly continuous, then can we not come up with a similar setting which will yield us new smooth functions? It turns out that the answer to this question is affirmative and it is given by a corollary of the next theorem:

Theorem 1.2.3.0.1. Let $n \in \mathbb{N}$ and $f, g \in L^1(\mathbb{R}^n)$ such that $\partial_i g$ exists and is bounded for all $i = 1, 2, \dots, n$. Then, $f * g \in C^1(\mathbb{R}^n)$ satisfying

$$\partial_i(f * g) = f * (\partial_i g).$$

Proof. The main idea of the proof involves the usage of DCT to a well defined function. Let $x \in \mathbb{R}^n$ be fixed. Define for $h \neq 0$,

$$\begin{aligned} \Phi_h : \mathbb{R}^n &\rightarrow \mathbb{R} \\ y &\mapsto f(y) \left| \frac{g(x - y + h e_i) - g(x - y)}{h} \right|. \end{aligned}$$

Then, for all $y \in \mathbb{R}^n$,

$$1. \lim_{h \rightarrow 0} \Phi_h(y) = f(y) \partial_i g(x - y).$$

2. By LMVT and the fact that each partial derivative of g is bounded, say, by a constant c , we have

$$\begin{aligned} |\Phi_h(y)| &= |f(y)| \left| \frac{g(x - y + he_i) - g(x - y)}{h} \right| \\ &= |f(y)| |\partial_i g(x - y + \theta e_i)| \\ &\leq c |f(y)|. \end{aligned}$$

Since, $f \in L^1(\mathbb{R}^n)$, from DCT it follows that

$$\lim_{h \rightarrow 0} \int_{\mathbb{R}^n} \Phi_h(y) dy = \int_{\mathbb{R}^n} f(y) \partial_i g(x - y) dy$$

and this implies

$$\lim_{h \rightarrow 0} \frac{1}{h} [f * g(x + he_i) - f * g(x)] = (f * \partial_i g)(x),$$

that is $\partial_i(f * g)$ exists and $\partial_i(f * g) = f * (\partial_i g)$. □

1.2.4 Smooth functions

Now, the corollary that we've been talking about before is as follows:

Corollary 1.2.4.0.1. *Let $n \in \mathbb{N}$ and let $f \in L^1(\mathbb{R}^n)$ and $g \in C_c^\infty(\mathbb{R}^n)$. Then, $f * g \in C^\infty(\mathbb{R}^n)$ and for all $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, we have*

$$\partial^\alpha(f * g) = f * \partial^\alpha g$$

in $\mathbb{R}^n$, that is,

$$\partial_1^{\alpha_1} \partial_2^{\alpha_2} \dots \partial_n^{\alpha_n} (f * g) = f * (\partial_1^{\alpha_1} \partial_2^{\alpha_2} \dots \partial_n^{\alpha_n} g).$$

Proof. This can be easily proved by applying induction on the partial differentials in the above proof. □

Chapter 2

Smooth Functions

In the previous chapter we've seen how to construct a new smooth function, given any two specific classes of functions. In this chapter, we'll learn some more ways of doing so and will also learn some of their nice properties. Construction of smooth functions are very important not only in analysis, but also, at times, in arenas such as differential geometry. We've already mentioned some of their importances in the preface. Another one would be when we are constructing smooth differential forms, which are nothing but smooth alternating multilinear functions (since, these are multilinear, the type of function being constructed is different, but the idea behind the construction remains almost same), wherein, we need the help of something called "auxiliary forms", which are compactly supported smooth functions, satisfying some nice properties. In our terminology, we'll call them *mollifiers*. Mollifiers are smooth functions with special properties used to create sequences of smooth functions approximating non-smooth functions, via convolution.

2.1 Mollifiers in $\mathbb{R}^n$

Let $n \in \mathbb{N}$. Define $\eta_0 \in C_c^\infty(\mathbb{R}^n)$ by

$$\eta_0(x) = \begin{cases} c \exp\left(\frac{1}{\|x\|^2-1}\right) & \text{if } \|x\| < 1, \\ 0 & \text{otherwise.} \end{cases}$$

where $c > 0$ is such that $\int_{\mathbb{R}^n} \eta_0 = 1$.

Definition 2.1.0.0.1. The standard mollifiers in $\mathbb{R}^n$ is the family $\{\eta_\epsilon\}_{\epsilon>0}$ in $C_c^\infty(\mathbb{R}^n)$ defined as

$$\eta_\epsilon(x) = \frac{1}{\epsilon} \eta_0\left(\frac{x}{\epsilon}\right)$$

for all $x \in \mathbb{R}^n$.

Remark 2.1.0.0.2. Note that the family of mollifiers $\{\eta_\epsilon\}_{\epsilon>0}$ satisfies

1. $0 \leq \eta_\epsilon(x) \leq \frac{c}{\epsilon^n}$ for all $x \in \mathbb{R}^n$.
2. $\text{supp}(\eta_\epsilon) = B[0, \epsilon] = \overline{B(0, \epsilon)}$.
3. $\int_{\mathbb{R}^n} \eta_\epsilon = 1$.

Definition 2.1.0.0.3. (L_{loc}^1 spaces). Let $n \in \mathbb{N}$ and $U \subseteq \mathbb{R}^n$ be open. Then,

$$L_{loc}^1(U) = \{f : U \rightarrow \overline{\mathbb{R}} \text{ measurable} | f \in L^1(K) \text{ for every compact subset } K \text{ of } U\}.$$

Remark 2.1.0.0.4. From the definition of L^1_{loc} spaces, it is an easy observation that $C(U) \subset L^1_{loc}(U)$.

Now, define

$$f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$x \mapsto \begin{cases} f(x) & \text{if } x \in U, \\ 0 & \text{otherwise.} \end{cases}$$

Note that if $f \in L^1(U)$, then, $f_0 \in L^1(\mathbb{R}^n)$, because the integral of f_0 over $\mathbb{R}^n$ reduces down to the integral of f over U . But, the converse is not true. It can be easily seen by considering the function $f(x) = \frac{1}{x}$.

2.1.1 Mollification of functions

Definition 2.1.1.0.1. (ϵ -mollification). Let $n \in \mathbb{N}$ and $U \subseteq \mathbb{R}^n$ be open and let $u \in L^1_{loc}(U)$. The ϵ -mollification of u is defined as

$$u_\epsilon(x) := (\eta_\epsilon * u_0)(x),$$

for all $x \in \mathbb{R}^n$, where $u_0 : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ is defined as

$$u_0(x) = \begin{cases} u(x) & \text{if } x \in U, \\ 0 & \text{otherwise.} \end{cases}$$

Remark 2.1.1.0.2. 1. Note that if $u \in L^1(U)$, then, $u_0 \in L^1(\mathbb{R}^n)$, then by corollary 1.2.3.0.1, it follows that $u_\epsilon(x) \in C^\infty(\mathbb{R}^n)$. In particular, since u_ϵ is a smooth function on a compact region, it implies that u_ϵ is bounded, that is, finite for all $x \in \mathbb{R}^n$.

2. Note that

$$\begin{aligned} u_\epsilon(x) &= \int_{\mathbb{R}^n} u_0(y) \eta_\epsilon(x - y) dy \\ &= \int_{B(x, \epsilon) \cap U} u(y) \eta_\epsilon(x - y) dy. \end{aligned}$$

Therefore,

(a) for all $x \notin B(U, \epsilon) := \{x \in \mathbb{R}^n | d(x, U) < \epsilon\}$, we have $B(x, \epsilon) \cap U = \emptyset$, and hence, in this case $u_\epsilon(x) = 0$.

(b) if we define

$$U_\epsilon := \{x \in U | d(x, \partial U) > \epsilon\}.$$

then, for all $x \in U_\epsilon$, we have $B[x, \epsilon] \subset U$, and thus,

$$\begin{aligned} u_\epsilon(x) &= \int_{\mathbb{R}^n} u(y) \eta_\epsilon(x - y) dy \\ &= \int_{B[x, \epsilon]} u(y) \eta_\epsilon(x - y) dy, \end{aligned}$$

that is,

$$|u_\epsilon(x)| \leq \int_{B[x, \epsilon]} |u(y)| \eta_\epsilon(x - y) dy \leq \frac{c}{\epsilon^n} \int_{B[x, \epsilon]} |u(y)| dy < \infty,$$

since, $u \in L^1_{loc}(U)$.

Hence, we have that $u_\epsilon(x) < \infty$ when either $u \in L^1(U)$ or $x \in U_\epsilon$, or $x \notin B(U, \epsilon)$. Also, note that for any compact subset $K \subset U$, for small enough ϵ , we have $K \subset U_\epsilon$.

2.1.2 Lebesgue points

Definition 2.1.2.0.1. (*Lebesgue point*). Let $n \in \mathbb{N}$ and $U \subseteq \mathbb{R}^n$ be open and $f \in L^1_{loc}(U)$. Then, $x_0 \in U$ is said to be a Lebesgue point of f if

$$\lim_{r \rightarrow 0} \frac{1}{\mathcal{L}^n(B(x_0, r))} \int_{B(x_0, r)} |f(y) - f(x_0)| dy = 0.$$

So, if f is a continuous function, then, every point of U is a Lebesgue point.

Furthermore, we define

$$L(f) := \{x_0 \in U \mid x_0 \text{ is a Lebesgue point of } f\}.$$

Note that the Lebesgue measure of a unit radius ball centered at the origin in $\mathbb{R}^n$ is given by the following formula:

$$W_n := \mathcal{L}^n(B(0, 1)) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$

W_n is maximum when $n = 5$ and $\lim_{n \rightarrow \infty} W_n = 0$.

By the fact that $\mathcal{L}^n(B(x_0, r)) = \mathcal{L}^n(B(0, 1))r^n$, the above formula is useful in determination of Lebesgue points of any function $f \in L^1_{loc}(U)$.

Theorem 2.1.2.0.2. (*Lebesgue differentiation theorem*). Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ be open and $f \in L^1_{loc}(U)$. Then, $\mathcal{L}^n(\mathbb{R}^n \setminus L(f)) = 0$.

2.2 Convergence of ϵ -mollification of functions

in this section, we will see three different types of convergence of ϵ -mollification of functions, namely, pointwise convergence, uniform convergence and L^p convergence.

2.2.1 Pointwise convergence in L^1_{loc} functions at Lebesgue points

Theorem 2.2.1.0.1. (*Pointwise convergence of the mollification sequence*). Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ be open, $f \in L^1_{loc}(U)$ and $x_0 \in L(f)$. Then,

$$\lim_{\epsilon \rightarrow 0} f_\epsilon(x_0) = f(x_0).$$

Proof. For $x_0 \in L(f)$, we can find $\epsilon_0 > 0$ such that $B[x_0, \epsilon_0] \in U$. Then, for all $\epsilon \in (0, \epsilon_0)$,

$$\begin{aligned} f_\epsilon(x_0) - f(x_0) &= \eta_\epsilon * f_0(x_0) - f(x_0) \\ &= \int_{B[x_0, \epsilon]} f(y) \eta_\epsilon(x_0 - y) dy - f(x_0) \int_{B[x_0, \epsilon]} \eta_\epsilon(x_0 - y) dy \\ &= \int_{B[x_0, \epsilon]} \eta_\epsilon(x_0 - y) [f(y) - f(x_0)] dy. \end{aligned}$$

(In the above equations, the second equality holds because $\int_{\mathbb{R}^n} \eta_\epsilon = 1$, and, $\text{supp } \eta_\epsilon = B[x_0, \epsilon]$.)

Then, for all $\epsilon \in (0, \epsilon_0)$,

$$\begin{aligned}
 |f_\epsilon(x_0) - f(x_0)| &\leq \int_{B[x_0, \epsilon_0]} \eta_\epsilon(x_0 - y) |f(y) - f(x_0)| dy \\
 &\leq \frac{c}{\epsilon^n} \int_{B(x_0, \epsilon_0)} |f(y) - f(x_0)| dy \\
 &= \frac{cW_n}{\epsilon^n} \times \frac{1}{\mathcal{L}^n(B(x_0, \epsilon_0))} \int_{B(x_0, \epsilon_0)} |f(y) - f(x_0)| dy \\
 &= 0.
 \end{aligned}$$

The last equality follows from the fact that x_0 is Lebesgue point. $\square$

Remark 2.2.1.0.2. From the above two theorems, it follows that if for $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ is open, $f \in L^1_{loc}(U)$ and $x_0 \in L(f)$, then,

$$\lim_{\epsilon \rightarrow 0} f_\epsilon = f, \text{ a.e. in } U.$$

An application - FT of Calculus of Variations

Theorem 2.2.1.0.3. (Fundamental theorem of Calculus of Variations). Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ be open, and $f \in L^1_{loc}(U)$ be such that

$$\int_U f \phi = 0, \text{ for all } \phi \in C_c^\infty(U), \text{ } —(*)$$

then, $f = 0$ a.e. in U .

Proof. It is enough to prove that $f(x) = 0$ for all $x \in L(f)$.

Let $x_0 \in L(f)$ be fixed. Then, there exists some $\epsilon_0 > 0$ such that $B[x_0, \epsilon_0] \subset U$. Then, for all $\epsilon \in (0, \epsilon_0)$, define

$$\begin{aligned}
 \phi_\epsilon : U &\longrightarrow \overline{\mathbb{R}} \\
 x &\longmapsto \eta(x_0 - x)
 \end{aligned}$$

where η_ϵ is the ϵ -mollification of η . Then, we have that $\text{supp}(\phi_\epsilon) = B[x_0, \epsilon] \subset B[x_0, \epsilon_0] \subset U$, that is, $\phi_\epsilon \in C_c^\infty(U)$, for all $\epsilon \in (0, \epsilon_0)$. So, from (*) it follows that for all $\epsilon \in (0, \epsilon_0)$,

$$\begin{aligned}
 \int_U f(y) \phi_\epsilon(y) dy &= 0, \\
 \text{or, } \int_U f(y) \eta_\epsilon(x_0 - y) dy &= 0, \\
 \text{or, } \int_{\mathbb{R}^n} f_0(y) \eta_\epsilon(x_0 - y) dy &= 0, \\
 \text{or, } (f_0 * \eta_\epsilon)(x_0) &= 0, \\
 \text{or, } f_\epsilon(x_0) &= 0.
 \end{aligned}$$

Now, since, $\lim_{\epsilon \rightarrow 0} f_\epsilon(x_0) = f(x_0)$, we have $f(x_0) = 0$. $\square$

Corollary 2.2.1.0.4. Let f, ϕ be same as in the above theorem and let $g \in L^1_{loc}(U)$ be such that

$$\int_U f \phi = \int_U g \phi,$$

then, $f = g$ a.e. in U .

2.2.2 Uniform convergence in continuous functions on compact subsets

We have already seen that when $f \in L^1_{loc}(U)$, for an open subset $U \subseteq \mathbb{R}^n$, then, the ϵ -mollification (f_ϵ) converges pointwise to f at the Lebesgue points of f . Now, instead of $f \in L^1_{loc}(U)$, if we have that $f \in C(U)$, then we have something better and it is given by the following theorem:

Theorem 2.2.2.0.1. (Uniform convergence of the mollification sequence). *Let $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$ be open and let $f \in C(U)$, then, for every compact $K \subset U$, (f_ϵ) converges to f in $C(K)$.*

Proof. Since, K is compact, $\exists$ some $\epsilon_0 > 0$ such that $B[K, \epsilon_0] \subset U$.

Let $\rho > 0$. Now, since f is continuous, f is uniformly continuous on $B[K, \epsilon_0]$, that is, $\exists \delta > 0$ such that for all $x, y \in B[K, \epsilon_0]$,

$$\|x - y\| < \delta \implies |f(x) - f(y)| < \rho.$$

Note that, for all $x \in K$ and $0 < \epsilon < \min\{\epsilon_0, \delta\}$, from theorem 2.2.1.0.1, it follows that

$$\begin{aligned} |f_\epsilon(x) - f(x)| &\leq \int_{B(x, \epsilon_0)} \eta_\epsilon(x - y) |f(y) - f(x)| dy \\ &< \rho \int_{B(x, \epsilon_0)} \eta_\epsilon(x - y) dy \\ &= \rho, \end{aligned}$$

that is, for all $0 < \epsilon < \min\{\epsilon_0, \delta\}$, $\|f_\epsilon - f\|_{C(K)} < \epsilon$. Since, ρ is independent of any $x \in K$, $f_\epsilon \rightarrow f$ in $C(K)$. $\square$

2.2.3 Convergence of L^p_{loc} functions on compact sets

Theorem 2.2.3.0.1. (Convergence of the mollification sequence in L^p). *Let $n \in \mathbb{N}$, $1 \leq p < \infty$, $U \subset \mathbb{R}^n$ be open and $f \in L^p_{loc}(U)$. The, for any compact set $K \subset U$, (f_ϵ) converges to f in $L^p(K)$.*

Proof. Let $K \subset U$ be a fixed compact set. By local compactness of $\mathbb{R}^n$, we can choose $\epsilon_0 > 0$ and an open set $V \subset\subset U$ such that $K \subset B[K, \epsilon_0] \subset V \subset\subset U$, where the notation $V \subset\subset U$ denotes that V is a pre-compact open subset of U . Using this, we will prove the theorem in two steps:

Step 1: We claim that for all $\epsilon \in (0, \epsilon_0)$,

$$\|f_\epsilon\|_{L^p(K)} \leq \|f\|_{L^p(V)}. \quad (2.1)$$

Note that for all $\epsilon \in (0, \epsilon_0)$ and $x \in K$,

$$f_\epsilon(x) = \int_{B(x, \epsilon)} \eta_\epsilon(x - y) f(y) dy.$$

Let q be the conjugate exponent of p , that is $\frac{1}{p} + \frac{1}{q} = 1$, then using H older's inequality,

$$\begin{aligned} |f_\epsilon(x)| &\leq \int_{B(x, \epsilon)} \eta_\epsilon(x - y) |f(y)| dy \\ &= \int_{B(x, \epsilon)} \eta_\epsilon^{1/p}(x - y) \eta_\epsilon^{1/q}(x - y) |f(y)| dy \\ &\leq \left(\int_{B(x, \epsilon)} \eta_\epsilon(x - y) dy \right)^{1/q} \left(\int_{B(x, \epsilon)} \eta_\epsilon(x - y) |f(y)|^p dy \right)^{1/p} \\ &= \left(\int_{B(x, \epsilon)} \eta_\epsilon(x - y) |f(y)|^p dy \right)^{1/p}. \end{aligned}$$

Raising to the power of p both sides, we get

$$|f_\epsilon(x)|^p \leq \left(\int_{B(0,\epsilon)} \eta_\epsilon(y) |f(x-y)|^p dy \right).$$

Integrating both sides wrt x and using Tonelli, we get that for all $\epsilon \in (0, \epsilon_0)$,

$$\begin{aligned} \|f_\epsilon\|_{L^p(K)}^p &\leq \int_{B(0,\epsilon)} \left(\int_K \eta_\epsilon(y) |f(x-y)|^p dx \right) dy \\ &= \int_{B(0,\epsilon)} \eta_\epsilon(y) \left(\int_{K-y} |f(z)|^p dz \right) dy, \end{aligned}$$

which we obtained from a change of variable $x - y = z$, and therefore, we have

$$\|f_\epsilon\|_{L^p(K)}^p \leq \|f\|_{L^p(V)}^p \int_{B[0,\epsilon]} \eta_\epsilon(y) dy = \|f\|_{L^p(V)}^p,$$

which proves our claim.

Step 2: Let $\rho > 0$ be given. By density of compactly supported continuous functions in L^p , we can choose a function $g \in C(\bar{V})$ such that $\|f - g\|_{L^p(V)} < \rho$. Therefore, for all $0 < \epsilon < \epsilon_0$,

$$\begin{aligned} \|f_\epsilon - f\|_{L^p(K)} &\leq \|f_\epsilon - g_\epsilon\|_{L^p(K)} + \|g_\epsilon - g\|_{L^p(K)} + \|g - f\|_{L^p(K)} \\ &\leq \|f - g\|_{L^p(V)} + M\|g_\epsilon - g\|_{C(K)} + \|g - f\|_{L^p(V)} \\ &= 2\|f - g\|_{L^p(V)} + M\|g_\epsilon - g\|_{C(K)}, \end{aligned}$$

where M is a constant. Taking the limit $\epsilon \rightarrow 0$ both sides, we have

$$\lim_{\epsilon \rightarrow 0} \|f_\epsilon - f\|_{L^p(K)} \leq 2\rho,$$

and thus our required convergence follows. $\square$

Theorem 2.2.3.0.2. *Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ and $f \in L^1_{loc}(U)$. then, for each $\epsilon > 0$, $f_\epsilon \in C^\infty(U_\epsilon)$ with*

$$\partial^\alpha f_\epsilon = (\partial^\alpha \eta_\epsilon) * f. \quad (2.2)$$

Furthermore, if $f \in L^1(U)$, then, $f_\epsilon \in C^\infty(\mathbb{R}^n)$ satisfying (2.2).

Proof. Clearly, if $f \in L^1(U)$, then, this theorem directly follows from corollary 1.2.4.0.1 by taking $g = \eta_\epsilon$.

When $f \in L^1_{loc}(U)$, $f \in L^1(K)$ for every compact subset $K \subset U$. Note that, for each $x \in U_\epsilon$,

$$f_\epsilon(x) = (\eta_\epsilon * f_0)(x) = \int_{B[x,\epsilon]} \eta_\epsilon(x-y) f_0(y) dy,$$

and since $B[x, \epsilon]$ is compact, it implies that for each $x \in U_\epsilon$, f_ϵ is defined.

Now, if we can establish 2.2, then, it will automatically imply $f_\epsilon \in C^\infty(U_\epsilon)$. But, proving 2.2 is not a difficult job, as we can follow the similar lines as the proof of 1.2.3.0.1, by defining $h = \epsilon$ and again as before, $g = \eta_\epsilon$. The only difference will be that the function Φ and the convolution integral involved in the proof of the aforementioned theorem is defined over $\mathbb{R}^n$, but in this case, they will reduce down to a function over U and an integral over $B[x, \epsilon]$, for each $x \in U_\epsilon$. Also, DCT will still hold in this case because the domain of the convolution integral is $B[x, \epsilon]$, so, we just need the dominating function f to be in $L^1(B[x, \epsilon])$ (that is, not necessarily $f \in L^1(U_\epsilon)$), for each $x \in U_\epsilon$, and it is indeed the case. Continuing this proof with induction over the partial differentials, we will obtain our required result 2.2. $\square$

Theorem 2.2.3.0.3. (Existence of cut-off function). *Let $n \in \mathbb{N}$, $K, U \subseteq \mathbb{R}^n$ with K compact, U open and $K \subset U$. Then, there exists $\phi \in C_c^\infty(\mathbb{R}^n)$ such that:*

1. $0 \leq \phi \leq 1$ in $\mathbb{R}^n$,
2. $\phi = 1$ on K ,
3. $\text{supp } \phi \subset U$.

Proof. Choose $\epsilon > 0$ such that $B[K, 3\epsilon] \subset U$.

Define

$$\begin{aligned} \phi : \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto (\mathcal{X}_{B[K, \epsilon]} * \eta_\epsilon)(x), \end{aligned}$$

that is, for all $x \in \mathbb{R}^n$,

$$\phi(x) = \int_{\mathbb{R}^n} (\mathcal{X}_{B[K, \epsilon]} \eta_\epsilon(x - y)) dy.$$

So, for all $x \in \mathbb{R}^n$, $\phi(x) \leq \int_{\mathbb{R}^n} \eta_\epsilon(x - y) dy = 1$, and since, by definition, the function ϕ is non-negative, we have $0 \leq \phi \leq 1$.

Now, for all $x \in K$,

$$\phi(x) = \int_{\mathbb{R}^n} \mathcal{X}_{B[K, \epsilon]}(y) \eta_\epsilon(x - y) dy = \int_{B[K, \epsilon]} \eta_\epsilon(x - y) dy = \int_{B(x, \epsilon)} \eta_\epsilon(x - y) dy = 1,$$

and therefore, $\phi(x) = 1$, for all $x \in K$.

To prove (3), let $x \in \mathbb{R}^n$ such that $\phi(x) \neq 0$. Then, there exists $y \in \mathbb{R}^n$ such that $\mathcal{X}_{B[K, \epsilon]}(y) \eta_\epsilon(x - y) \neq 0$, that is, $\exists y \in B[K, \epsilon]$ such that $x - y \in B[0, \epsilon]$. So,

$$x = y + (x - y) \in B[K, \epsilon] + B[0, \epsilon] = B[K, 2\epsilon],$$

and thus, $\text{supp } \phi \subseteq B[K, 2\epsilon] \subset U$. □

Theorem 2.2.3.0.4. *Let $n \in \mathbb{N}$, $C \subset \mathbb{R}^n$ be closed. Then, there exists $f \in C^\infty(\mathbb{R}^n)$ such that:*

1. $0 \leq f \leq 1$,
2. $f^{-1}(0) = C$.

Proof. For any closed set C , we can find a sequence of compact sets $\{K_m\}_{m \in \mathbb{N}}$ such that

$$\mathbb{R}^n \setminus C = \bigcup_{m \in \mathbb{N}} K_m \text{ and } K_m \subset K_{m+1} \text{ for all } m \in \mathbb{N}.$$

Let us see why.

Now, by the previous theorem, for each $m \in \mathbb{N}$, we can find $f_m \in C_c^\infty(\mathbb{R}^n)$ such that

1. $0 \leq f_m \leq 1$ in $\mathbb{R}^n$,
2. $f_m = 1$ on K_m ,
3. $\text{supp } f_m \subset \mathbb{R}^n \setminus C$.

Define

$$\begin{aligned} f : \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto \sum_{m \in \mathbb{N}} \frac{f_m(x)}{2^m(1 + \|f_m\|_\infty)}. \end{aligned}$$

We show that $f \in C^\infty(\mathbb{R}^n)$. Note that

$$|f(x)| = \left| \sum_{m \in \mathbb{N}} \frac{f_m(x)}{2^m(1 + \|f_m\|_\infty)} \right| \leq \sum_{m \in \mathbb{N}} \frac{1}{2^m} = 1.$$

Hence, by Weierstrass M-test, it follows that the series of functions involved in the definition of f is uniformly convergent, and therefore, term-by-term differentiation of this series wrt x is valid. Clearly, the only factor depending on x in each term of the series is $f_m(x)$, which for each $m \in \mathbb{N}$ is infinitely differentiable. Thus, we can differentiate the series term-by-term infinitely many times; that is, we can differentiate f infinitely many times. Hence, f is a smooth function.

It is easy to see that $f^{-1}(0) = C$ because in $\mathbb{R}^n$, $f \equiv 0$ iff $f_m = 0$ for each $m \in \mathbb{N}$, and since, $\text{supp } f_m \subset \mathbb{R}^n \setminus C$, $f_m = 0$ for each $m \in \mathbb{N}$ in $\mathbb{R}^n$. $\square$

Theorem 2.2.3.0.5. *Let $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$ be open and let $1 \leq p < \infty$. Then, $C_c^\infty(U)$ is dense in $L^p(U)$.*

Proof. Let us find a sequence of open sets $\{V_m\}_{m \in \mathbb{N}}$ such that for all $m \in \mathbb{N}$, we have

1. $U = \bigcup_{m \in \mathbb{N}} V_m$,
2. $\overline{V_m}$ is compact,
3. $\overline{V_m} \subset V_{m+1}$,
4. $B[V_m, \frac{1}{m}] \subset U$.

We can always choose such open sets V_m by considering

$$V_m = \{x \in U : d(x, \partial U) > 1/m\} \bigcap B(0, m),$$

for all $m \in \mathbb{N}$. Defining $S_m = \{x \in U : d(x, \partial U) > 1/m\}$, note that S_m itself satisfies all, but (2), the above conditions of V_m . For (2) to hold, we need the intersection of S_m with $B(0, m)$ for each $m \in \mathbb{N}$. To show the density of $C_c^\infty(U)$ in $L^p(U)$, we first consider any arbitrary function $f \in L^p(U)$.

Define

$$\begin{aligned} \phi_m : \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto \chi_{V_m}(x)f_0(x). \end{aligned}$$

Note that, $|\phi_m(x)| \leq |f_0(x)|$, where, $f_0 \in L^p(\mathbb{R}^n)$. Also,

$$\phi_m(x) - f_0(x) = f(x)\chi_{V_m}(x) - f(x) = f(x)(\chi_{U \setminus V_m}(x)),$$

which for $x \in U$, tends to 0 as $m \rightarrow \infty$, and for $x \notin U$, $f(x) = 0$, which implies $\phi_m(x) - f_0(x) = 0$. Therefore, by DCT, $\phi_m \rightarrow f_0$ in $L^p(\mathbb{R}^n)$. So, we can find $M \in \mathbb{N}$ such that $\|\phi_M - f_0\|_{L^p(\mathbb{R}^n)} < \epsilon/2$.

Define

$$\begin{aligned} g_k : \mathbb{R}^n &\longrightarrow \mathbb{R} \\ x &\longmapsto (\phi_M * \eta_{\frac{1}{M+k+1}})(x). \end{aligned}$$

We have,

$$g_k(x) = \int_{B[x, \frac{1}{M+k+1}] \cap V_M} \mathcal{X}_{V_M}(y) f_0(y) \eta_{\frac{1}{M+k+1}}(x-y) dy$$

Using the fact that if $f \in L_c^p(U)$ and $\eta \in C_c^\infty(U)$, then, $f * \eta \in C_c^\infty(U)$, we get that $g_k \in C_c^\infty(U)$. By convergence on compact subsets in L^p spaces, we can find $k_0 \in \mathbb{N}$ such that

$$\|\phi_M - g_{k_0}\|_{L^p(U)} = \|\phi_M - g_{k_0}\|_{L^p(B[V_M, 1/M])} < \epsilon/2.$$

Using triangle inequality,

$$\|f - g_{k_0}\|_{L^p(U)} \leq \|f - \phi_M\|_{L^p(U)} + \|g_{k_0} - \phi_M\|_{L^p(U)} < \epsilon.$$

□

2.3 Partition of Unity

In this section, we will show how we can obtain two different types of partition of unity wrt any given open cover, say, $\{U_\alpha\}$ of an open subset $U \subseteq \mathbb{R}^n$. In the first case, the partition of unity will belong to the class of C_c^∞ , whereas in the second case, the partition of unity that we will get is subordinate to the open cover.

2.3.1 Partition of Unity I

Theorem 2.3.1.0.1. *Let $n \in \mathbb{N}$ and $U \subset \mathbb{R}^n$ be open and let $\{U_\alpha : \alpha \in I\}$ be an open cover for U . Then, there exists a family $\{\psi_m : m \in \mathbb{N}\}$ in $C_c^\infty(U)$ such that*

1. $0 \leq \psi_m \leq 1$ in U , for all $m \in \mathbb{N}$,
2. for each $m \in \mathbb{N}$, there exists $\alpha(m) \in I$ such that $\text{supp}(\psi_m) \subset U_{\alpha(m)}$,
3. for all $x \in U$, $\sum_{m \in \mathbb{N}} \psi_m(x) = 1$, and
4. for every compact set $K \subset \mathbb{R}^n$, there exists $l = l(K) \in \mathbb{N}$, an open set $V \subset\subset U$ such that $K \subset V$ and $\sum_{j=1}^l \psi_j(x) = 1$, for all $x \in V$.

Proof. Since, any open $U \subseteq \mathbb{R}^n$ is separable, we can choose a countable dense subset $D \subset U$. We consider the family

$$\mathfrak{F} := \{B[x, r] | x \in D, r \in \mathbb{Q} \cap (0, 1) \text{ and for some } \alpha \in I, B[x, r] \subset U_\alpha \cap U\}.$$

By definition, clearly $\mathfrak{F}$ is a non-empty countable set. Therefore, we can write

$$\mathfrak{F} = \{B[x_m, r_m] | m \in \mathbb{N}\}.$$

Now, note that by density of D in U ,

$$U \subseteq \bigcup_{m \in \mathbb{N}} B\left(x_m, \frac{r_m}{2}\right). \quad (2.3)$$

By theorem 2.2.3.0.3, we can find $\phi_m \in C_c^\infty(U)$ satisfying

1. $0 \leq \phi_m \leq 1$ in U
2. $\phi_m = 1$ on $B[x_m, \frac{r_m}{2}]$ and
3. $\text{supp } \phi_m \subset B(x_m, r_m)$.

We now construct $\{\psi_m | m \in \mathbb{N}\}$. For each $m \in \mathbb{N}$, we define $\psi_m \in C_c^\infty(U)$ by

$$\psi_1 = \phi_1 \text{ in } U, \quad \psi_{m+1} := \phi_{m+1} \prod_{j=1}^m (1 - \phi_j) \quad (2.4)$$

in U , for all $m \in \mathbb{N}$. Clearly, $0 \leq \psi_m \leq 1$, for all $m \in \mathbb{N}$, and that proves (1).

Furthermore, note that, for each $m \in \mathbb{N}$,

$$\text{supp } (\psi_m) \subset \text{supp } (\phi_m) \subset B(x_m, r_m).$$

Therefore, it follows from the definition of $\mathfrak{F}$ that for each $m \in \mathbb{N}$, there exists $\alpha(m) \in I$ such that $\text{supp } \psi_m \subset U_{\alpha(m)}$, which proves (2).

To prove (3), first note that by using induction on equation 2.4, we get

$$\sum_{j=1}^m \psi_j = 1 - \prod_{j=1}^m (1 - \phi_j) \quad (2.5)$$

in U . Hence, for all $m \in \mathbb{N}$,

$$\sum_{j=1}^m \psi_j(x) = 1, \text{ for all } x \in \bigcup_{j=1}^m B(x_j, r_j), \quad (2.6)$$

because there will be at least one j for which ϕ_j will be 1.

Now, let $a \in U$ be fixed. Then, using 2.3, there exists $p \in \mathbb{N}$ such that $a \in B(x_p, \frac{r_p}{2})$. From 2.6, it follows that $\sum_{j=1}^p \psi_j(a) = 1$ and for all $m \geq p$, $\psi_m(a) = 0$, which implies for all $m \geq p$, $\sum_{j=1}^m \psi_j(a) = 1$. Letting $m \rightarrow \infty$, we obtain (3).

It remains to prove (4). Let $K \subset U$ be compact. Using 2.3, it implies that we can find $l = l(K) \in \mathbb{N}$ such that

$$K \subseteq \bigcup_{m=1}^l B(x_m, \frac{r_m}{2}).$$

Let us set $V := \bigcup_{m=1}^l B(x_m, \frac{r_m}{2})$. Then, using 2.6, we see that

$$\sum_{j=1}^l \psi_j(x) = 1, \text{ for all } x \in V,$$

and thus, our proof is complete. □

2.3.2 Partition of Unity II

Theorem 2.3.2.0.1. *Let $n \in \mathbb{N}$, $U \subseteq \mathbb{R}^n$ be open and let $\{U_\alpha | \alpha \in I\}$ be an open cover for U . Then, there exists a family $\{\psi_\alpha | \alpha \in I\}$ in $C^\infty(U)$ such that*

1. $0 \leq \psi_\alpha \leq 1$ in U , for all $\alpha \in I$
2. for all $\alpha \in I$, $\text{supp } \psi_\alpha \subset U_\alpha$
3. $\{\text{supp } (\psi_\alpha) | \alpha \in I\}$ is a locally finite open cover for U .
4. $\sum_{\alpha \in I} \psi_\alpha = 1$ in U .

In this case, we say that $\{\psi_\alpha | \alpha \in I\}$ is a partition of unity subordinate to the open cover $\{U_\alpha | \alpha \in I\}$.

Proof. We consider two cases:

Case 1: Let us suppose the family $\mathfrak{F} := \{U_\alpha | \alpha \in I\}$ is locally finite.

Using theorem 2.3.1.0.1, we find $\{\phi_m | m \in \mathbb{N}\}$ in $C_c^\infty(U)$ such that

1. $0 \leq \phi_m \leq 1$ in U , for all $m \in \mathbb{N}$,
2. for each $m \in \mathbb{N}$, there exists $\alpha(m) \in I$ such that $\text{supp } (\phi_m) \subset U_{\alpha(m)}$,
3. for all $x \in U$, $\sum_{m \in \mathbb{N}} \phi_m(x) = 1$, and
4. for every compact set $K \subset \mathbb{R}^n$, there exists $l = l(K) \in \mathbb{N}$, an open set $V \subset\subset U$ such that $K \subset V$ and $\sum_{m=1}^l \phi_j(x) = 1$, for all $x \in V$.

For each $\beta \in I$, let us set

$$S_\beta := \{m \in \mathbb{N} | \alpha(m) = \beta\}.$$

Note that S_β may be empty and $\bigcup_{\beta \in I} S_\beta = \mathbb{N}$. Also, since support of ϕ_m , for each $m \in \mathbb{N}$, lies in a unique $U_{\alpha(m)}$, it implies for $\beta \neq \beta'$, $S_\beta \cap S_{\beta'} = \emptyset$.

Define for each $\alpha \in I$, $\psi_\alpha \in C^\infty(U)$ in U by

$$\psi_\alpha := \sum_{m \in S_\alpha} \phi_m.$$

If for any α , S_α is empty, then, $\psi_\alpha = 0$. Also, by the 4th property, note that for each compact set $K \subset U$, there exists a finite set $\mathbb{N}_k \subset \mathbb{N}$ such that $\phi_m = 0$ for all $m \in \mathbb{N} \setminus \mathbb{N}_k$.

Now, by construction, for some α , it might happen that S_α is infinite, in which case the corresponding ψ_α will then be an infinite sum, so there's a possibility that ψ_α might not be in C^∞ , because infinite sum need not preserve smoothness. But, it is enough to show that for each $x \in U$, there exists a nbd $V_x \subset\subset U$, such that $\psi \in C^\infty(V_x)$. From the 4th property listed above and the fact that $\{x\}$ is a compact set in $\mathbb{R}^n$, for each $\alpha \in I$, $x \in V_x$, $\psi_\alpha(x)$ becomes a finite sum of smooth functions which is smooth, thus implying the family $\{\psi_\alpha | \alpha \in I\}$ lies in $C^\infty(U)$.

Keeping in mind all of these, let us now prove (1), (2), (3) and (4).

(1) follows from the above listed 3rd property and that $S_\alpha \subset \mathbb{N}$, for each $\alpha \in I$.

(2) follows from the definition of ψ_α , for each $\alpha \in I$.

For (3), first note that

$$\text{supp } (\psi_\alpha) = \bigcup_{m \in S_\alpha} \text{supp } (\phi_m).$$

Also, $U \subseteq \bigcup_{\alpha \in I} \text{supp } \psi_\alpha$, because for each $x \in U$, there exists $\alpha \in I$ such that $x \in \text{supp } \psi_\alpha$. This can be seen by the definition of ψ_α and the fact that $\sum_{m \in \mathbb{N}} \phi_m(x) = 1$, for each $x \in U$. Since, $\{U_\alpha\}$ is a locally finite open cover and for all $m \in S_\alpha$, $\text{supp } \phi_m \subset U_\alpha$, that is, $\text{supp } \psi_\alpha \subseteq U_\alpha$, it follows that $\{\text{supp } (\psi_\alpha) : \alpha \in I\}$ is a locally finite open cover for U .

Now, for $\alpha \neq \alpha'$, since, $S_\alpha \cap S_{\alpha'} = \emptyset$ and $\mathbb{N} = \bigcup_{\alpha \in I} S_\alpha$, we have

$$\sum_{\alpha \in I} \psi_\alpha = \sum_{\alpha \in I} \sum_{m \in S_\alpha} \phi_m = 1,$$

thus, (4) follows.

Case 2: Suppose the family $\mathfrak{F} := \{U_\alpha | \alpha \in I\}$ is not locally finite.

By the precompactness property of $\mathbb{R}^n$, we can obtain a locally finite refinement $\{V_\beta | \beta \in J\}$ of the open cover $\{U_\alpha | \alpha \in I\}$. Now, by case 1, we can find a partition of unity $\{\phi_\beta | \beta \in J\}$ subordinate to the family $\mathfrak{G} := \{V_\beta | \beta \in J\}$. Since, $\mathfrak{G}$ is a refinement of the open cover $\mathfrak{F}$, it follows that for each $\beta \in J$, there exists $f(\beta) \in I$ such that $V_\beta \subset U_{f(\beta)}$, where $f : J \rightarrow I$ is an onto function.

Define $\psi_\alpha \in C^\infty(U)$ such that

$$\psi_\alpha(x) = \sum_{\substack{\beta \in J \\ f(\beta) = \alpha}} \phi_\beta(x),$$

for all $x \in U$.

We show that $\{\psi_\alpha | \alpha \in I\}$ forms a partition of unity subordinate to the open cover $\{U_\alpha | \alpha \in I\}$, that is, it satisfies the properties (1), (2), (3) and (4) of the theorem.

For (1), there's nothing to prove, as by definition of ψ_α , it follows that for each $\alpha \in I$, $0 \leq \psi_\alpha \leq 1$.

For (2), note that, for each $\alpha \in I$,

$$\text{supp } \psi_\alpha = \bigcup_{\substack{\beta \in J \\ f(\beta) = \alpha}} \text{supp } (\phi_\beta) \subseteq U_\alpha.$$

We now prove (3). Let $x \in U$ be arbitrary. Since, $\{\phi_\beta | \beta \in J\}$ is a partition of unity subordinate to an open cover of U , we have that $\{\text{supp } (\phi_\beta) | \beta \in J\}$ is a locally finite open cover of U . Thus, given any nbd, say, U_x of x , there exists only finitely many $\beta \in J$ such that $\text{supp } (\phi_\beta)$ intersects U_x . Hence, from (2), it follows that for all $x \in U$, $\{\text{supp } \psi_\alpha | \alpha \in I\}$ forms a locally finite open cover of U .

Lastly, to prove (4), observe that if $\beta, \beta' \in J$ such that $f(\beta) = \alpha$ and $f(\beta') = \alpha'$, for some $\alpha, \alpha' \in I$ where $\alpha \neq \alpha'$, then, $\text{supp } (\phi_\beta) \cap \text{supp } (\phi_{\beta'}) = \emptyset$. Therefore,

$$\sum_{\alpha \in I} \psi_\alpha = \sum_{\alpha \in I} \sum_{\substack{\beta \in J \\ f(\beta) = \alpha}} \phi_\beta = \sum_{\beta \in J} \phi_\beta = 1.$$

□

2.3.3 Smooth Urysohn's lemma

Lemma 2.3.3.0.1. *Let $n \in \mathbb{N}$ and let A, B be two non-empty disjoint closed subsets of $\mathbb{R}^n$. then, there exists $f \in C^\infty(\mathbb{R}^n)$ such that $f(A) = \{0\}$ and $f(B) = \{1\}$.*

Proof. This lemma is an easy application of partition of unity. Note that $\{\mathbb{R}^n \setminus A, \mathbb{R}^n \setminus B\}$ is an open cover for $\mathbb{R}^n$. Let $\{\psi_1, \psi_2\}$ be a partition of unity subordinate to this open cover, that is,

1. $0 \leq \psi_1, \psi_2 \leq 1$ in $\mathbb{R}^n$,
2. $\text{supp } (\psi_1) \subseteq \mathbb{R}^n \setminus A$ and, $\text{supp } (\psi_2) \subseteq \mathbb{R}^n \setminus B$,
3. $\psi_1 + \psi_2 = 1$ in $\mathbb{R}^n$.

Define $f \in C^\infty(\mathbb{R}^n)$ by $f = \psi_1$. Using the above two properties, it is an easy observation that this function f is our required function. $\square$

Part II

Differentiation Theory

Chapter 3

Signed Measures

Till date, the measure theory that we have learnt, we have always assumed the codomain of a measure to be the non-negative real line. Signed measures give a generalization to the concept of measures in a sense that, the codomain of a signed measure is allowed to be the extended real or the extended complex line. If we recall, our main motto of bringing in the entire concept of measure theory was to produce a version of the fundamental theorem of calculus for Lebesgue integrals, so as the chapter progresses, hopefully one will be able to relate how this entire process of generalizing measures is important. We would like to differentiate a measure ν with respect to another measure μ on the same σ -algebra, and such differentiation usually proceeds more naturally only in the general setting. After developing all the abstract theory, if we restrict μ to be the Lebesgue measure on $\mathbb{R}$, we will be able to obtain a refined result that would finally give the required version of FTC. There are also other applications of signed measures, for example, they can represent things such as electric charge that can be either positive or negative, and moreover, complex measures also have some functional-analytic significance.

Definition 3.0.0.0.1. Let $(X, \mathcal{M})$ be a measurable space. A signed measure is a function $\nu : \mathcal{M} \rightarrow [-\infty, \infty]$ such that

1. $\nu(\emptyset) = 0$,
2. ν assumes at most one of the values $\pm\infty$,
3. ν is countably additive on disjoint sets.

The usual measures that we have dealt with before, we will call it as positive measures.

Remark 3.0.0.0.2. If $\{E_j\}$ is a collection of disjoint sets in $(X, \mathcal{M}, \nu)$, then, $\nu(\bigsqcup E_j) = \sum \nu(E_j)$, and if $\nu(\bigsqcup E_j)$ is finite, then, $\sum \nu(E_j)$ is absolutely convergent. Moreover, we have the following decomposition:

$$\sum \nu(E_j) = \sum_{\nu(E_j) > 0} \nu(E_j) + \sum_{\nu(E_j) < 0} \nu(E_j).$$

Lemma 3.0.0.0.3. Let $E, F \in \mathcal{M}$ such that $E \subseteq F$, then

1. if $\nu(E) \in \mathbb{R}$, then, $\nu(F \setminus E) = \nu(F) - \nu(E)$.
2. if $\nu(E) = \pm\infty$, then, $\nu(F) = \pm\infty$.

Remark 3.0.0.0.4. Let $(X, \mathcal{M}, \nu)$ be a measure space. Then, if there exists $E \in \mathcal{M}$ such that $\nu(E) = +\infty$, then, $\nu(X) = +\infty$, and there cannot exist any other $F \in \mathcal{M}$ such that $\nu(F) = -\infty$. So, every signed measure is either $\nu : \mathcal{M} \rightarrow (-\infty, \infty]$, or, $\nu : [-\infty, \infty)$.

Remark 3.0.0.0.5. The monotonicity property of measures, which holds in the case of positive measures, doesn't hold in the case of signed measures. For example, let $(\mathbb{R}, \mathcal{M}, \mu)$ be the Lebesgue measure space. For all $E \in \mathcal{M}$, we define

$$\nu(E) = \int_E f \, d\mu,$$

where f is any function integrable in the extended sense. Clearly, ν defines a signed measure on $\mathbb{R}$. Now, if we consider $f = \mathcal{X}_{[0,1]} - \mathcal{X}_{[2,3]}$, then, $\nu(\mathbb{R}) = 0 < 1 = \nu([0, 1])$. Hence, the monotonicity property fails.

Remark 3.0.0.0.6. The theorem for continuity from above and continuity from below holds in the case of signed measures.

Let $(X, \mathcal{M}, \nu)$ be a measure space, where ν is a signed measure.

Definition 3.0.0.0.7. (Positive set). A set $P \in \mathcal{M}$ is said to be a positive set if for all $E \subset P$, $E \in \mathcal{M}$ and $\nu(E) \geq 0$.

Definition 3.0.0.0.8. (Negative set). A set $N \in \mathcal{M}$ is said to be a negative set if for all $E \subset N$, $E \in \mathcal{M}$ and $\nu(E) \leq 0$.

Definition 3.0.0.0.9. (Null set). A set $F \in \mathcal{M}$ is said to be a null set if for all $G \subset F$, $G \in \mathcal{M}$ and $\nu(G) = 0$.

Lemma 3.0.0.0.10. Let $(X, \mathcal{M}, \nu)$ be a measure space, where ν is a signed measure. Then,

1. measurable subsets of a positive set is a positive set,
2. countable union of positive sets is a positive set.

Proof. (i) is obvious from definition of positive sets. For (ii), let us consider a countable collection of positive sets, say, $\{P_j\}$, then, we need to show that $\bigcup_j P_j$ is also a positive set. Converting this union into a disjoint union, it is easy to check that all the properties of a positive set follows. $\square$

Remark 3.0.0.0.11. The second part of the above lemma also holds for negative sets and null sets.

Lemma 3.0.0.0.12. Let $E \in \mathcal{M}$ be such that $0 < \nu(E) < \infty$. Then there exists a positive set $\tilde{E} \subset E$ with $\nu(\tilde{E}) > 0$.

Proof. If E is a positive set, then we are done. Suppose not, then, there exists a set $E_1 \in \mathcal{M}$ such that $E_1 \subseteq E$ with $\nu(E_1) < 0$. By Archimedian principle, there exists $n \in \mathbb{N}$ such that $\nu(E_1) < -\frac{1}{n}$. Consider the set

$$M_1 := \left\{ n : \exists A \in \mathcal{M} \text{ s.t. } A \subseteq E_1 \text{ and } \nu(A) < -\frac{1}{n} \right\},$$

By well ordering principle, there exists a least element of the set M_1 , say n_1 such that $\nu(E_1) < -\frac{1}{n_1}$. Then, $\nu(E \setminus E_1) > \nu(E) + \frac{1}{n_1} > 0$. If the set $E \setminus E_1$ is a positive set, then we are done. Otherwise, continuing by induction, there exists a set $E_k \subset E \setminus (\bigcup_{j=1}^{k-1} E_j)$ such that $\nu(E_k) < 0$. Considering the set

$$M_k := \left\{ n : \exists A \in \mathcal{M} \text{ s.t. } A \subseteq E_k \text{ and } \nu(A) < -\frac{1}{n} \right\},$$

there exists least element n_k such that $\nu(E_k) < -\frac{1}{n_k}$. If the induction process stops at a finite step, then, we will have our required positive set. If not, we will have a disjoint collection of negative sets $\{E_j\}$ such that $\bigcup_1^\infty E_k \subset E$ with $\sum_1^\infty \nu(E_k) < -\sum_1^\infty \frac{1}{n_k}$, that is, $\sum_1^\infty \frac{1}{n_k} < -\sum_1^\infty \nu(E_k) < \infty$. Thus, $n_k \rightarrow \infty$ as $k \rightarrow \infty$.

Let, $B := E \setminus \bigcup_1^\infty E_k$. We claim that B is a positive set. If not, there exists a smallest $n_0 \in \mathbb{N}$ and a set $E_0 \subset B$ such that $\nu(E_0) < -\frac{1}{n_0}$. But, as $n_k \rightarrow \infty$, there exists a n_j such that $n_0 < n_j$. Since, $E_0 \subset E \setminus \bigcup_1^\infty E_k \subset E \setminus \bigcup_1^{j-1} E_k$, it follows that n_0 is an element of M_j where $n_0 < n_j$. This contradicts the construction of n_j . Therefore, B is our required positive set. $\square$

3.1 Hahn decomposition theorem

Now, let us prove one of the most important theorems of this chapter.

Theorem 3.1.0.0.1. (Hahn decomposition theorem). *Let ν be a signed measure on a measurable space $(X, \mathcal{M})$. Then, there exists a positive set $P \in \mathcal{M}$ and a negative set $N \in \mathcal{M}$ such that $X = P \sqcup N$. Furthermore, if (P', N') is another such pair, then, $P \Delta P' (= N \Delta N')$ is a null set.*

Proof. Without loss of generality, let us consider the signed measure $\nu : \mathcal{M} \rightarrow [-\infty, \infty)$ (otherwise, consider $-\nu$). Let us define

$$S = \{E : E \in \mathcal{M} \text{ is a positive set}\}.$$

Let, $\alpha = \sup_S \nu(E)$. Then, there exists a positive set P such that $\nu(P) = \alpha$. Indeed, there is a sequence $\{P_j\}$ of positive sets converging to α . Converting these sets into disjoint sets, say, $\{Q_j\}$ and considering their union $\bigcup_j Q_j$, it follows from continuity from below that $\bigcup_j Q_j$ is the required positive set P .

Let, $N = P^c$. We claim that N is a negative set. If not, there exists a set $C \subseteq N$ such that $\nu(C) > 0$. By the previous lemma, there exists a positive set $C' \subseteq C$. Since, C' and P are two disjoint positive sets of X , it follows that

$$\nu(C' \sqcup P) = \nu(C') + \nu(P) = \alpha + \nu(C') > \alpha,$$

and this contradicts the definition of P . and by definition of P and N , clearly, $X = P \sqcup N$.

Now, suppose there exists another pair (P', N') . Then,

$$P \Delta P' = (P \setminus P') \cup (P' \setminus P) = (N' \setminus N) \cup (N \setminus N') = N \Delta N'.$$

Also, since, $P \setminus P' = N' \setminus N$ is a subset of both P and N' , which are positive and negative sets respectively, it follows that both $P \setminus P'$ and $N' \setminus N$ are null sets. Similarly, $P' \setminus P$ and $N \setminus N'$ are also null sets. Thus, by definition, both $P \Delta P'$ and $N \Delta N'$ are null sets. $\square$

Remark 3.1.0.0.2. Suppose $(X, \mathcal{M}, \nu)$ is a measure space, where ν is a signed measure. Let, (P, N) be a Hahn decomposition of ν . Then, for any $E \in \mathcal{M}$, we can define

$$\begin{aligned} \nu^+(E) &= \nu(E \cap P) \geq 0, \\ \nu^-(E) &= -\nu(E \cap N) \geq 0. \end{aligned}$$

Therefore,

$$\begin{aligned} \nu(E) &= \nu(E \cap P) + \nu(E \cap N) \\ &= \nu^+(E) - \nu^-(E), \end{aligned}$$

that is, $\nu = \nu^+ - \nu^-$. **Thus, every signed measure can be written as the difference of two positive measures.** Moreover, this decomposition is unique and the two measures ν^+ and ν^- are mutually singular to each other.

Let us first define what we mean by mutually singular, then, the Jordan decomposition theorem will provide us with the uniqueness of this decomposition.

Definition 3.1.0.0.3. (Mutually singular). Two signed measures μ and ν are said to be mutually singular (or singular wrt each other), denoted as $\mu \perp \nu$, if there exists a partition of $X = E \sqcup F$, $E, F \in \mathcal{M}$, such that E is a null set for μ and F is a null set for ν .

3.2 Jordan decomposition theorem

Theorem 3.2.0.0.1. (Jordan decomposition theorem). *Let ν be a signed measure on a measurable space $(X, \mathcal{M})$. Then, there exists unique positive measures ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.*

Proof. From remark 3.1.0.0.2, we already have the existence of such mutually singular measures ν^+ and ν^- , so, we only need to show their uniqueness. Let us suppose ν_1 and ν_2 are two positive measures on $(X, \mathcal{M})$ such that $\nu = \nu_1 - \nu_2$ and $\nu_1 \perp \nu_2$. Thus, there exists $A, B \in \mathcal{M}$ such that $X = A \sqcup B$, such that A, B are null sets wrt ν_1 and ν_2 respectively. To begin, we first show that the pair (B, A) is indeed a Hahn decomposition for X . For any $E \in \mathcal{M}$,

$$\nu(E \cap B) = \nu_1(E \cap B) - \nu_2(E \cap B) = \nu_1(E \cap B) = \nu_1(E \cap B) + \nu_1(E \cap A) = \nu_1(E) > 0,$$

that, is, B is a positive set wrt ν . Similarly, it can be shown that $\nu(E \cap A) = -\nu_2(E)$, which implies that A is a negative set wrt ν . Also, since, $X = A \sqcup B$, it follows that (B, A) is a Hahn decomposition for X .

Now,

$$\nu_1(E) = \nu(E \cap B) = \nu(E \cap B \cap P) + \nu(E \cap B \cap N) = \nu(E \cap B \cap P) = \nu(E \cap B \cap P) + \nu(E \cap A \cap P) = \nu(E \cap P) = \nu^+(E).$$

Similarly, $\nu_2(E) = \nu^-(E)$. Therefore, the uniqueness of the measures ν^+ and ν^- follows. $\square$

3.3 Total variation measure of a signed measure

Let $(X, \mathcal{M})$ be a measurable space and ν be a signed measure defined on this space. By Jordan decomposition theorem, we have a decomposition of $\nu = \nu^+ + \nu^-$, where $\nu^+ \perp \nu^-$. The total variation measure of ν is $|\nu| = \nu^+ + \nu^-$. Note that, for any $E \in \mathcal{M}$,

$$|\nu(E)| \leq \nu^+(E) + \nu^-(E) = |\nu|(E).$$

Here, $|\nu|(E)$ is in fact the least positive measure dominating $|\nu(E)|$. It will be proved later at the time of discussing complex measures.

3.3.1 Important results

Lemma 3.3.1.0.1. *Let $(X, \mathcal{M})$ be a measurable space. Also, let ν be a signed measure and μ be a positive measure defined on $(X, \mathcal{M})$. Then,*

1. E is a null set for $\nu \iff \nu^+(E), \nu^-(E) = 0 \iff |\nu|(E) = 0$.
2. $\nu \perp \mu \iff \nu^+ \perp \mu, \nu^- \perp \mu \iff |\nu| \perp \mu$.

Proof. (1) Since, ν^+ and ν^- are mutually singular, there exists $A, B \in \mathcal{M}$ such that $X = A \sqcup B$, A is a null set for ν^- and B is a null set for ν^+ . Let, $E \in \mathcal{M}$ be a null set for ν .

Now,

$$\nu^+(E) = \nu^+(E \cap A) + \nu^+(E \cap B) = \nu^+(E \cap A) - \nu^-(E \cap A) = \nu(E \cap A) = 0.$$

Similarly, $\nu^-(E) = 0$. By definition of $|\nu|$, it also follows that $|\nu|(E) = 0$.

Converse of the statement, starting with $|\nu|(E) = 0$, follows from definitions.

(2) can also be easily proven by using (1) and playing with the definitions. $\square$

NOTE: In function spaces, it is enough to work with positive measures instead of signed measures. It is because for any $f \in L^1(\nu)$, $\int f d\nu = \int f d\nu^+ - \int f d\nu^-$, and therefore, $L^1(\nu) = L^1(\nu^+) \cap L^1(\nu^-)$. So, it follows that $L^1(\nu) = L^1(|\nu|)$.

3.3.2 Absolute continuity

Definition 3.3.2.0.1. Let μ be a positive measure and ν be a signed measure defined on a measurable space $(X, \mathcal{M})$. Then, ν is said to be absolute continuous wrt μ , denoted by $\nu \ll \mu$, if for any $E \in \mathcal{M}$, $\mu(E) = 0$ implies $\nu(E) = 0$.

Facts:

1. $\nu \ll \mu \iff \nu^+ \ll \mu \text{ and } \nu^- \ll \mu \iff |\nu| \ll \mu$.
2. $\nu \ll \mu \text{ and } \nu \perp \mu \implies \nu = 0$.

The following lemma can be considered to give a vague explanation of the nomenclature "absolute continuity".

Lemma 3.3.2.0.2. Let ν and μ be a signed measure and a positive measure respectively defined on a measurable space $(X, \mathcal{M})$. Then, $\nu \ll \mu$ iff given $\epsilon > 0$, there exists $\delta > 0$ such that $\mu(E) < \delta$ implies $|\nu(E)| < \epsilon$.

Proof. (TODO). □

Let $f \in L^1(\mu)$. Let us define

$$\nu(E) = \int_E f d\mu.$$

It is easy to observe that ν is a signed measure such that $\nu \ll \mu$. This gives rise to the following corollary.

Corollary 3.3.2.0.3. Let $f \in L^1(\mu)$. Then, given $\epsilon > 0$, there exists $\delta > 0$ such that $\mu(E) < \delta$ implies $|\int_E f d\mu| < \epsilon$.

The converse of this theorem also holds true, that is, given any positive measure μ and a signed measure ν , there exists a measurable integrable function f such that $\nu = \int f d\mu$. The proof of a stronger version of this converse statement, which also says about the uniqueness of the function f , is given by the following well known theorem, known as the Radon-Nikodym theorem.

3.4 Radon-Nikodym theorem

Theorem 3.4.0.0.1. (Radon-Nikodym theorem). Let $(X, \mathcal{M})$ be a measurable space. Let, μ be a σ -finite positive measure and ν be a σ -finite signed measure that takes value in $(-\infty, \infty]$. Then,

1. there exists unique σ -finite measures ν_a, ν_s such that $\nu = \nu_a + \nu_s$ such that $\nu_a \ll \mu$ and $\nu_s \perp \mu$.
2. there exists unique (up to μ measure zero sets) measurable integrable (in extended sense) function f such that $d\nu_a = f d\mu$.

Proof. (TODO). □

Remark 3.4.0.0.2. In the above theorem, σ -finiteness of the measures ν and μ is important, otherwise the theorem might not hold true. For example, consider the Lebesgue measure m and the counting measure c on $\mathbb{R}$. If m were to be absolutely continuous wrt c , then there should exist a measurable integrable function f such that $m(E) = \int_E f dc$, for all $E \in \mathcal{M}$. But, note that if we consider $E = \{x\}$, then, $m(E) = 0$, whereas since $\int_E f dc = f(x)$, we have $f(x) = 0$, for all $x \in \mathbb{R}$. This forces us to conclude that the Lebesgue measure is a zero measure, which is not true, hence contradiction. So, m cannot be absolutely continuous wrt c .

Now, suppose all the assumptions in the hypothesis of the Radon-Nikodyn theorem hold, then, for all $E \in \mathcal{M}$, there exists measurable integrable function f up to μ -measure zero such that

$$\nu(E) = \int_E f \, d\mu,$$

that is, for any other measurable integrable function g , we have

$$\int_E g \, d\mu = \int_E fg \, d\mu.$$

Now, we can write $\nu^+(E) = \int_E f^+ \, d\mu$ and $\nu^-(E) = \int_E f^- \, d\mu$, hence, we have

$$\int_E g \, d\nu^+ = \int_E gf^+ \, d\mu.$$

Now, with this in mind, let us define the Radon-Nikodyn derivative.

3.4.1 The Radon-Nikodyn derivative

Let ν , μ and f be same as defined before, then, we have $d\nu = f d\mu$, that is,

$$\frac{d\nu}{d\mu} = f,$$

and then, f is called the *Radon-Nikodyn derivative of ν wrt μ* .

For σ -finite signed measures ν_1 and ν_2 that are absolutely continuous wrt μ , we have

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu},$$

and, there exists f_1 and f_2 such that

$$\nu_1(E) = \int_E f_1 \, d\mu, \quad \nu_2(E) = \int_E f_2 \, d\mu.$$

Thus, we have

$$(\nu_1 + \nu_2)(E) = \int_E (f_1 + f_2) \, d\mu.$$

Proposition 3.4.1.0.1. *Let ν be a σ -finite signed measure and μ, λ be σ -finite positive measures such that $\nu \ll \mu$ and $\mu \ll \lambda$. Then,*

1. *if $g \in L^1(\nu)$, then, $g \left(\frac{d\nu}{d\mu} \right) \in L^1(\mu)$ and $\int g \, d\nu = \int g \left(\frac{d\nu}{d\mu} \right) \, d\mu$.*
2. *$\nu \ll \lambda$ and $\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda}$ λ a.e.*

Corollary 3.4.1.0.2. *If $\mu \ll \lambda$ and $\lambda \ll \mu$, then, $\frac{d\lambda}{d\mu} \frac{d\mu}{d\lambda} = 1$ a.e. (with respect to either λ or μ).*

Chapter 4

Complex measures

A complex measure on a measurable space $(X, \mathcal{M})$ is a function $\nu : \mathcal{M} \rightarrow \mathbb{C}$ such that

1. $\nu(\phi) = 0$, and,
2. if $\{E_j\}$ is sequence of disjoint sets in $\mathcal{M}$, then, $\nu(\cup_1^\infty E_j) = \sum_1^\infty \nu(E_j)$, where the series converges absolutely.

As complex numbers can be broken down into its real and imaginary parts, complex measure, say ν can also be broken down into its real and imaginary parts, say ν_r and ν_i respectively, where ν_r and ν_i are signed measures that do not assume the values $\pm\infty$. So, the range of every complex measure is a bounded subset of $\mathbb{C}$.

Radon-Nikodym theorem for complex measures

Theorem 4.0.0.0.1. *Let $(X, \mathcal{M})$ be a measurable space. Let, μ be a σ -finite positive measure and ν be a complex measure. Then,*

1. *there exists unique complex measures ν_a, ν_s such that $\nu = \nu_a + \nu_s$ such that $\nu_a \ll \mu$ and $\nu_s \perp \mu$.*
2. *there exists unique (up to μ measure zero sets) $f \in L^1(\mu)$ such that $d\nu_a = f d\mu$.*

4.1 Total variation of a complex measure

Let ν be a complex measure on a measurable space $(X, \mathcal{M})$, then, total variation of ν is the positive measure $|\nu|$ such that $d\nu = f d\mu$ for any positive measure μ and $f \in L^1(\mu)$ implies $d|\nu| = |f|d\mu$.

Note that such a positive measure μ always exists by taking $\mu = |\nu_r| + |\nu_i|$, which is a σ -finite positive measure, and thus, by Radon-Nikodym theorem for complex measures, we also have the existence of $f \in L^1(\mu)$ such that $d\nu = f d\mu$. Now, to show the uniqueness of f and μ in the definition of $|\nu|$, let us suppose there exists f_1, f_2 and μ_1, μ_2 such that $d\nu = f_1 d\mu_1$ and $d\nu = f_2 d\mu_2$ for $f_1 \in L^1(\mu_1)$, $f_2 \in L^1(\mu_2)$. Let $\rho = \mu_1 + \mu_2$. Clearly, μ_1 and μ_2 are absolutely continuous wrt ρ . Hence, by proposition 3.4.1.0.1,

$$f_1 \frac{d\mu_1}{d\rho} d\rho = d\nu = f_2 \frac{d\mu_2}{d\rho} d\rho,$$

which implies

$$f_1 \frac{d\mu_1}{d\rho} = f_2 \frac{d\mu_2}{d\rho} \quad \rho \text{ a.e.}$$

Since, $\frac{d\mu_1}{d\rho}$ and $\frac{d\mu_2}{d\rho}$ are non-negative, it follows that

$$|f_1| \frac{d\mu_1}{d\rho} = \left| f_1 \frac{d\mu_1}{d\rho} \right| = \left| f_2 \frac{d\mu_2}{d\rho} \right| = |f_2| \frac{d\mu_2}{d\rho},$$

and therefore,

$$|f_1| d\mu_1 = |f_1| \frac{d\mu_1}{d\rho} d\rho = |f_2| \frac{d\mu_2}{d\rho} d\rho = |f_2| d\mu_2.$$

Hence, the definition of $|\nu|$ is independent of the choice of μ and f .

Also, note that definition of $|\nu|$ agrees with the definition of total variation of a signed measure, for in that case if ν is a signed measure, then, there exists a Hahn decomposition of $X = P \sqcup N$. Thus, we can write $d\nu = (\mathcal{X}_P - \mathcal{X}_N) d|\nu|$ by taking the function $f = \mathcal{X}_P - \mathcal{X}_N$ and $|\nu|$ as the positive measure, and it holds because $|\mathcal{X}_P - \mathcal{X}_N| = 1$. Then, integrating over any $E \in \mathcal{M}$ we obtain our definition of total variation of a signed measure, that is, $|\nu| = |\nu^+| + |\nu^-|$.

Proposition 4.1.0.0.1. *Let ν be a complex measure on $(X, \mathcal{M})$. Then,*

1. $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$.
2. $\nu \ll |\nu|$ and $\frac{d\nu}{d|\nu|}$ has absolute value 1 $|\nu|$ a.e.
3. $L^1(\nu) = L^1(|\nu|)$ and if $f \in L^1(\nu)$, then, $|\int f d\nu| \leq \int |f| d|\nu|$.

Proposition 4.1.0.0.2. *If ν_1, ν_2 are complex measures on $(X, \mathcal{M})$, then, $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$.*

Remark 4.1.0.0.3. $|\nu|$ is a finite measure. Indeed, for any $E \in \mathcal{M}$, we have

$$\nu(E) = \nu_r^+(E) - \nu_r^-(E) + i\nu_i^+(E) - i\nu_i^-(E),$$

that is,

$$|\nu(E)| = |\nu_r^+(E)| + |\nu_r^-(E)| + |\nu_i^+(E)| + |\nu_i^-(E)|.$$

Since, all of $|\nu_r^+(E)|, |\nu_r^-(E)|, |\nu_i^+(E)|, |\nu_i^-(E)|$ are finite measures and by definition, for all $E \in \mathcal{M}$, $|\nu|(E)$ is the smallest positive measure dominating $|\nu(E)|$, it follows that $|\nu|$ is a finite measure.

4.2 The space of complex measures

Let $(X, \mathcal{M})$ be a measurable space. Let us define

$$M(X) = \{\text{all complex measures on } X\}.$$

It is not difficult to see that $M(X)$ forms a vector space. Indeed, the zero measure is present as the identity element, the addition of two complex measures is a complex measure, and also the product of two complex measures is a complex measure. We define the following norm on the space $(M(X), \|\cdot\|)$: for all $\nu \in M(X)$,

$$\|\nu\| = |\nu|(X).$$

The positivity and homogeneity property of $\|\cdot\|$ is easy to see. The triangle inequality, after expanding the definition of $\|\cdot\|$, follows from proposition 4.1.0.0.2. Thus, $\|\cdot\|$ indeed defines a norm on $M(X)$.

Theorem 4.2.0.0.1. *$(M(X), \|\cdot\|)$ is a Banach space.*

Proof. Choose a Cauchy sequence $\{\nu_n\}$ in $M(X)$. Then, for all $E \in \mathcal{M}$, $\{\nu_n(E)\}$ is a Cauchy sequence in $\mathbb{C}$, which is a Banach space. Let us suppose

$$\lim_{n \rightarrow \infty} \nu_n(E) = \nu(E).$$

If we can show that ν is a complex measure, that is, if we can show that ν satisfies the subadditivity property, we are done. Let $\{E_n\}$ be a sequence of disjoint sets in X . We need to show that $\nu(\cap_1^\infty E_n) = \sum_1^\infty \nu(E_n)$.

Claim: If $F_n \in \mathcal{M}$ is a decreasing sequence of disjoint sets such that $\cap_1^\infty F_n = \phi$, then, $\nu(F_n) \rightarrow 0$ as $n \rightarrow \infty$.

Note that the above claim is enough to prove the required, as if we consider $E = \cup_1^\infty E_n$ and $F_n = E \setminus (\cup_1^n E_n)$, the above claim implies

$$\left| \nu(E) - \sum_1^n \nu(E_n) \right| \rightarrow 0 \text{ as } n \rightarrow \infty,$$

that is, $\nu(\sqcup_1^\infty E_n) = \sum_1^\infty \nu(E_n)$.

Also, the proof of the claim follows by using the Cauchy property of the sequence $\{\nu_n\}$ and the fact that elements of this sequence are functions of bounded variation, that is, their total variation is bounded. $\square$